

# AI for Transportation: From concepts to implementation

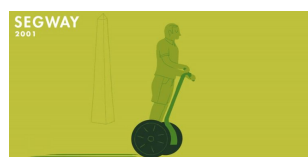
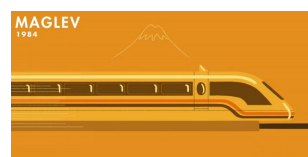
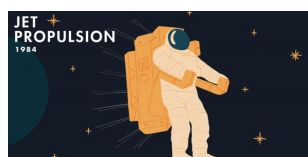
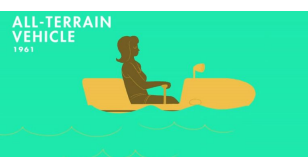
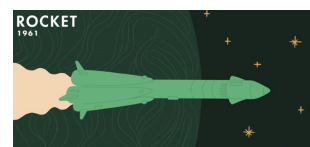
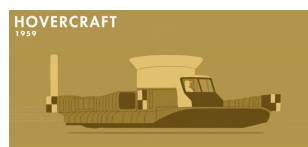
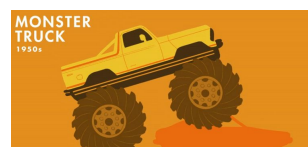
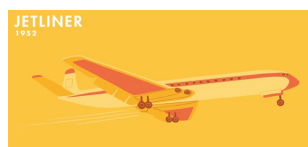
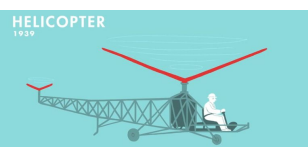
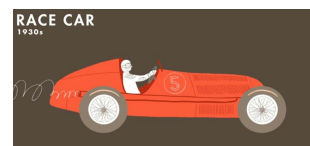
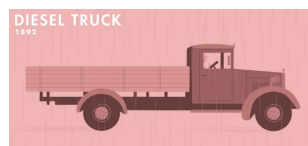
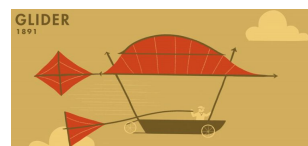
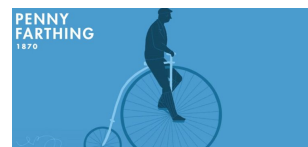
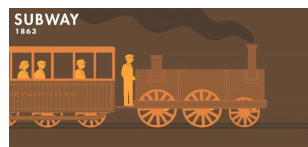
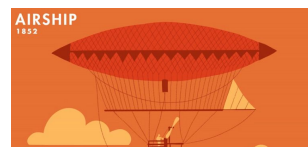
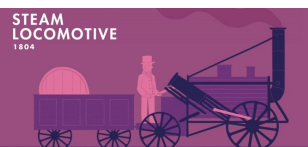
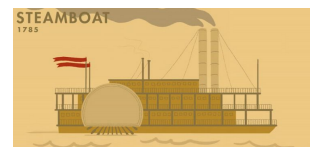
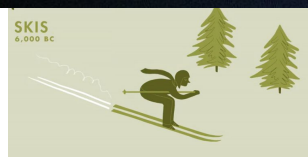


## Lecture 1: Introduction



What is success?

What defines the future?





Morning traffic on the Southeast Expressway in Dorchester. (David L. Ryan/Globe Staff)





Traffic in Beijing today





Beijing 1982 | Source: Wang Wenlan, Wen's Lens, <https://tinyurl.com/34kparne>





# Three questions for the students



Q1:

How many people were killed in road traffic crashes in the US in 2021?



42,915



42,915

0



electric  
vehicles



autonomous  
vehicles



road side  
robotics



e-scooters  
e-bikes



urban  
drones



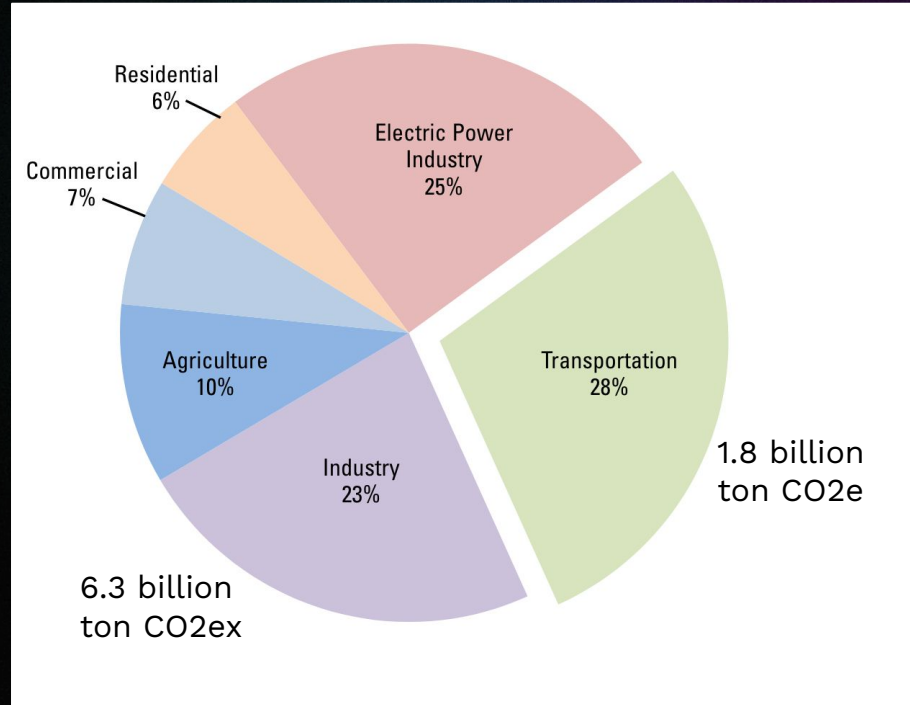


Q2:

How much does transportation contribute to the GHG emission in the US?



# U.S. GHG Emissions by Economic Sector 2022



**US: 28%**

**vs**

**Global: 16%**



Q3:

Is public transit affordable in Boston?



# Are we successful?

- Cities are heavily congested
- 42,915 killed on US roads
- 28% of GHG emission
- Many cannot afford public transit in Boston



Technology progress has **NOT** been translated to the betterment of the mobility system.



# What will define the future?



# Behavior + Computation



# P1. Behavioral thinking

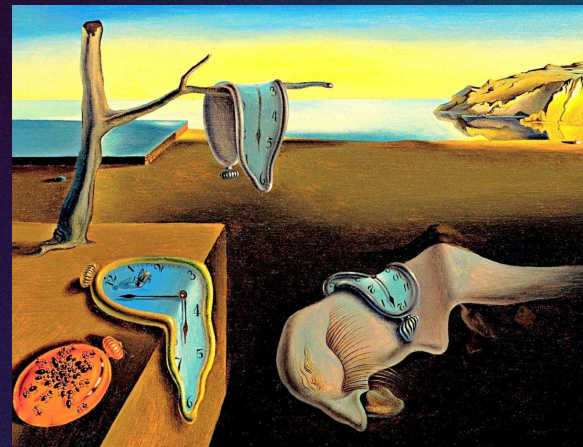
is travel social?



is travel emotional?



is time absolute?



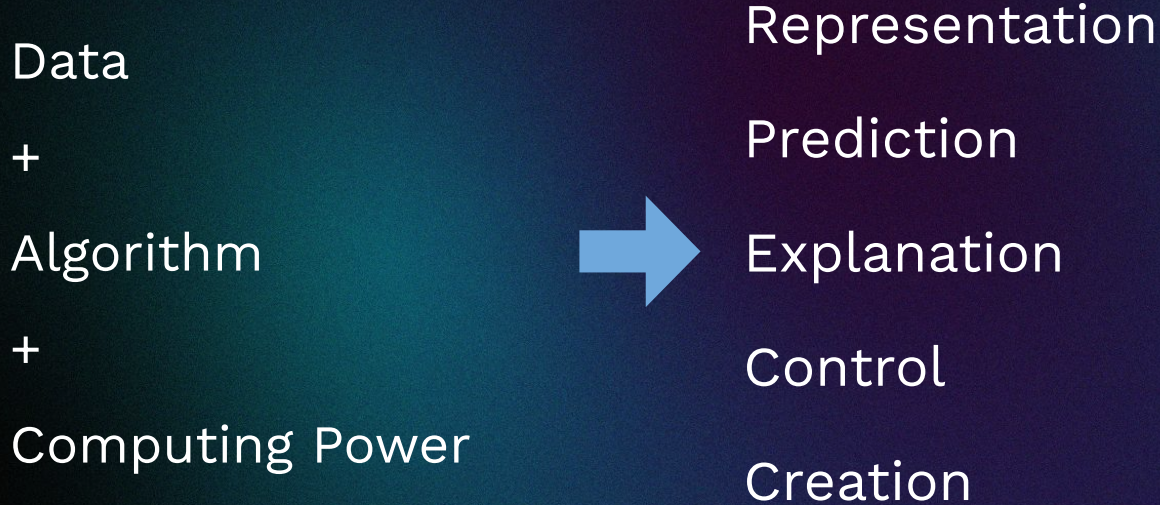


**Laziness knows no boundaries ...**





# P2: Computational Thinking





## Behavioral Thinking

- Emotional
- Social
- Perceptual



## Transportation Technology

- Electrification
- Automation
- Connectivity
- Sharing

---

## Computational Foundation

- Representation
- Explanation
- Prediction
- Control
- Creation



# AI for Transportation

L1: Introduction

L2: AI Models and Approaches

L3: Real-World AI Case Studies

L4: Generative AI



# AI for Transportation

## L1: Introduction

S1: Success and Future

S2: Transportation: System, Changes and Data

S3: Grounded AI: Structure, Trend, Why Now?

S4: AI: Tensions and Opportunities



# S2: Transportation Systems



# What is transportation?



The movement of people and goods



The collection of stakeholders – individuals and institutions – that deliver mobility for people

# What is the Transportation System?

## Physical Framework

### Vehicles



### Energy



### Infrastructure



## Functional Framework



### Technological



### Behavioral



### Political / Regulatory



### Financial



### Environmental

## Stakeholder Framework

### Business

- Traditional Mobility (Boeing, Toyota, DHL)
- Big Tech (Alphabet/Waymo, Amazon)
- Related Industries: Energy, Finance, Finance, Insurance, Infrastructure, etc.
- Startups and New Entrants

### Government

- International Governance (e.g., ICAO)
- National
- State
- Municipal

### Civil Society

- NGOs (Greenpeace, road safety, etc)
- IOs (e.g., WBCSD, WEF, World Bank)
- Academic/Research



# What is changing in Transportation?



# What is changing in Transportation?

## Technology

- Automation
- Electrification
- 5G/Connected
- Shared economy
- ...

## Data

- Ubiquitous sensing
- AI / computing
- Cybersecurity
- Property right
- ...

## Value

- Climate change
- Future of work
- Public health
- Social justice
- Urban livability
- ...



# Different framings of transportation

As a congestion problem

As a sustainability problem

As a social justice problem

As a personal identity problem

As an urban creativity problem

As a public health problem



# What is Data in Transportation?



Case study #1:

# Evolution of Data Analytics in Transport for London





- LTDS / CRM
- Oyster / Octopus
- iBus
- Mobile CDR / Wifi / RFID / Bluetooth
- Tracking Apps: MOVES / FMS / Strava
- Mobile App
- Text: Incidence Log, Customer Feedback, Twitter, youtube...



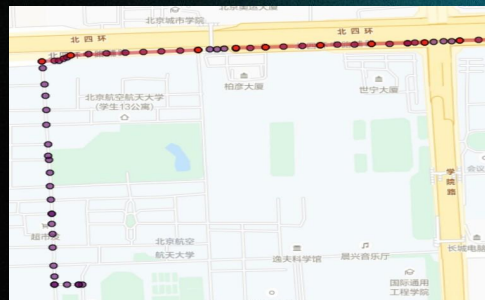
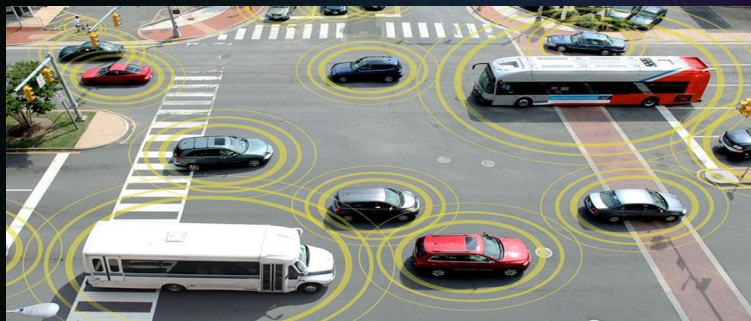
# Data Resources

- Location data, Trajectory data
- Transaction data
- Profile data
- Sensors: multimedia data
- Cross-platform identification



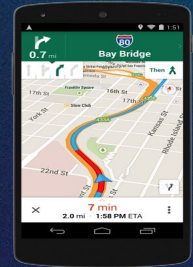
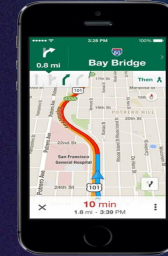


# Location Data and Floating-Car Trajectory

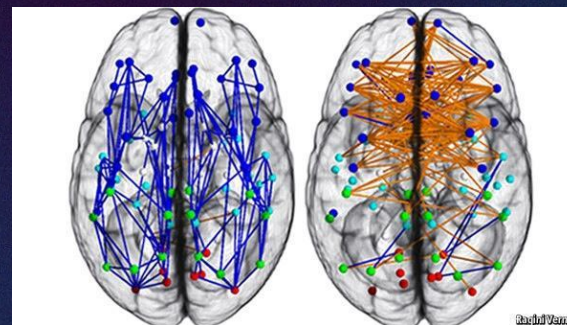
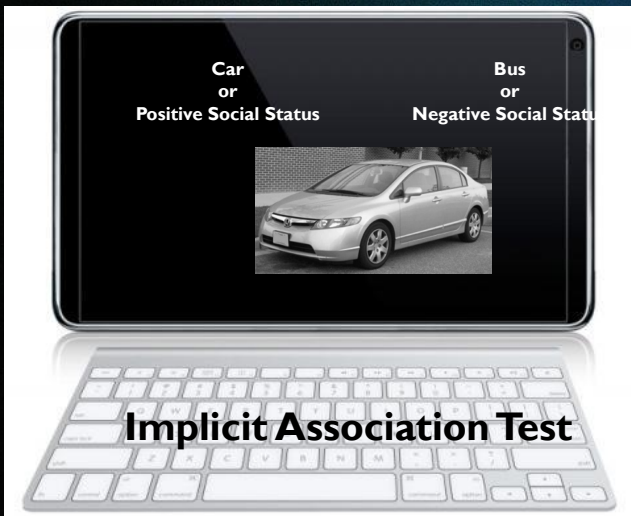




# Sensors: Loop detector, camera, microphone, mobile sensors ...









# Multi-modal Data Fusion and Analytics

**Multimodal Travel  
Information Platform**

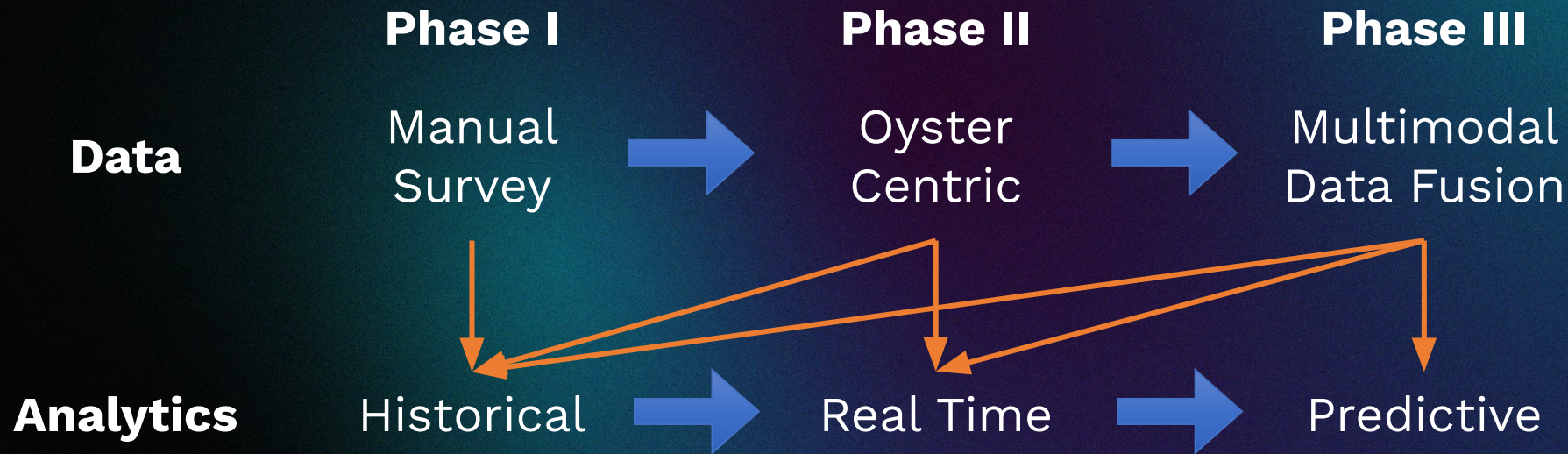
+

**Statistical Data  
Fusion**

- Oyster
  - Mobile CDR / Wifi / RFID / Bluetooth
  - LTDS / CRM
  - Tracking Apps: MOVES / FMS
- 
- Regression
  - Discrete Choice Modeling
  - Machine Learning (Unsupervised, supervised, reinforcement)
  - LLM and Foundational Models



# TfL Data Analytics





# Multi-channel view of cities



# Plato: Allegory of the Cave



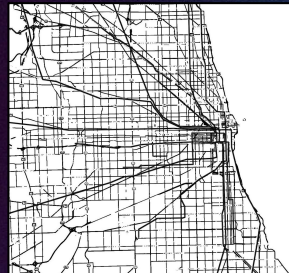


# A multi-channel view city (e.g. Chicago)

## Numeric values

Population: 2,693,976  
Auto ownership per HH:  
1.12/HH  
...

## Graphs (Transit and Road)



## Images (Aerial or Street-view)



## Natural Languages (From Wikipedia)

Chicago, officially the City of Chicago, is the most populous city in the U.S. state of Illinois, and the third-most-populous city in the United States. With an estimated population of 2,693,976 in 2019, it is also the most populous city in the Midwestern United States. Chicago is the county seat of Cook County, the second-most-populous county in the US, with a small portion of the northwest side of the city extending into DuPage County near O'Hare ...



# A multi-channel view city (e.g. Chicago)

Cities projected to **numbers**



Cities projected to **graphs**



Cities projected to **images**



Cities projected to **natural language**





# DNN: A unified framework for Graphs, Language, Images and Numbers

$$P(y_i \mid x_i) = f(x_i)$$

$$P(y_i \mid x_i, \text{Image}, \text{Text}, \text{Graph}) = f(x_i, \text{Image}, \text{Text}, \text{Graph}) \quad [\text{GLIN}]$$

Diagram illustrating the unified framework for Graphs, Language, Images and Numbers (GLIN). The input  $x_i$  is a vector representation that can be derived from various data types: Images, Language, and Numbers. The output  $y_i$  is a probability distribution over the target space, which can be a class label (e.g., "We love Chicago") or a graph structure (e.g., a triangle). The framework is labeled [GLIN].



# S3: Grounded AI in Transportation

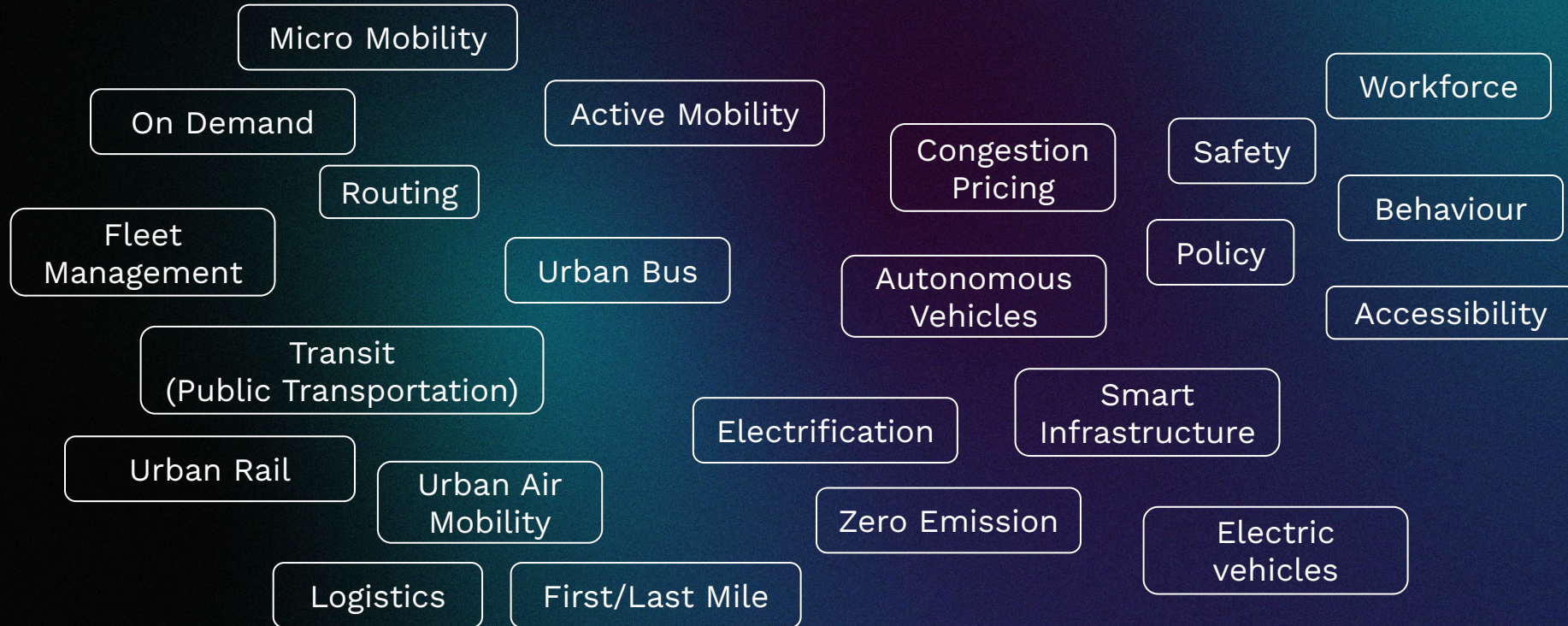


# Outline

- Wide range of transportation topics: how to structure them?
- Leading trends in mobility industry
- AI in transportation – why now?
- What's Grounded AI?



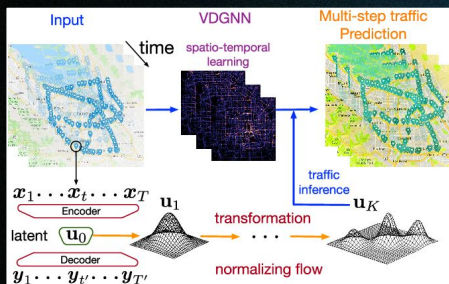
# The wide range transportation topics



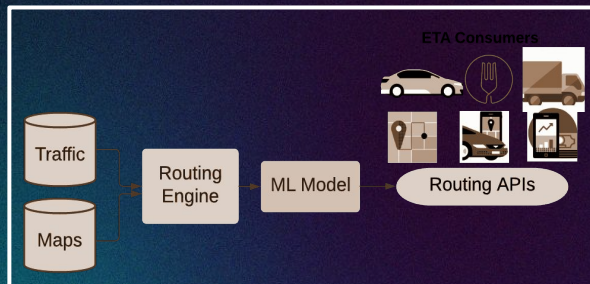


# AI-Driven Urban Mobility Management

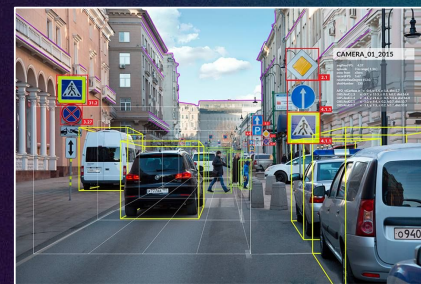
## Traffic Management



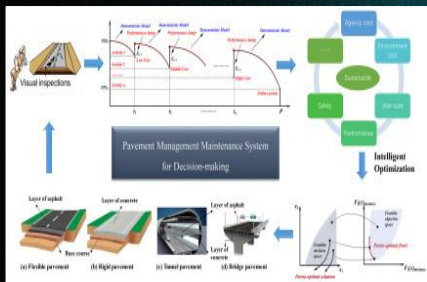
## On-Demand Mobility Operation



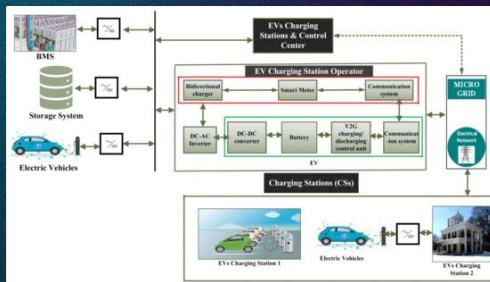
## Assisted or Automated Driving



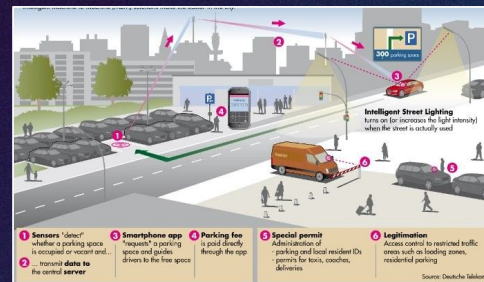
## Proactive Infrastructure Maintenance



## EV Charging/Discharging Management

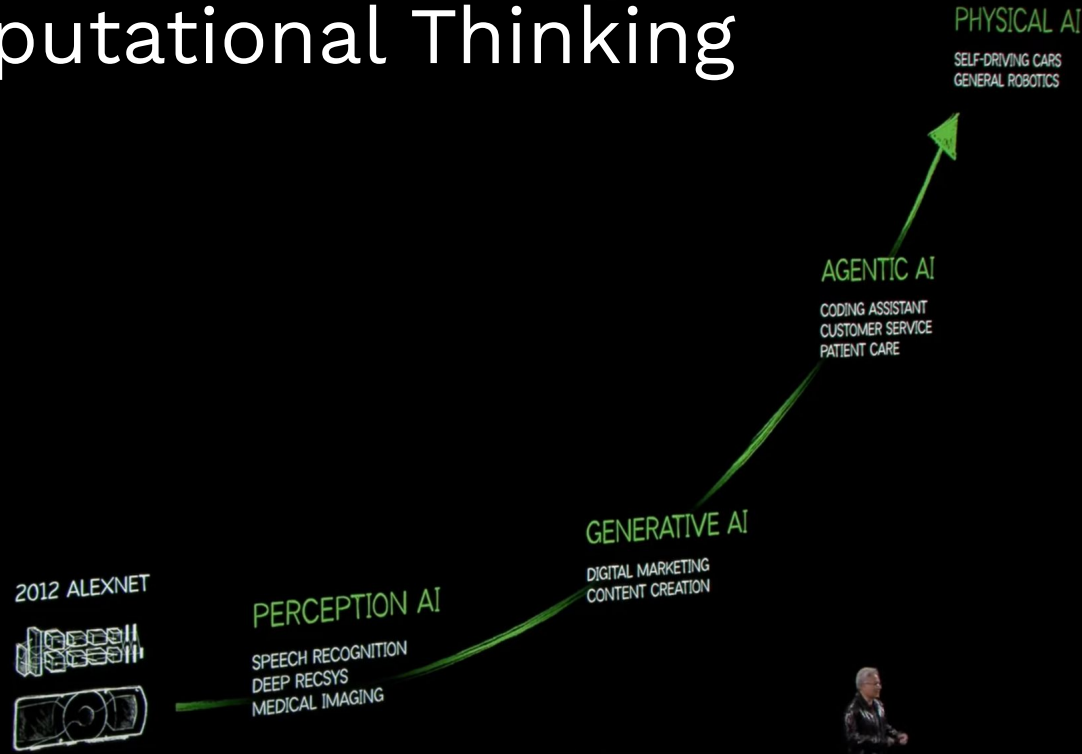


## Smart Parking



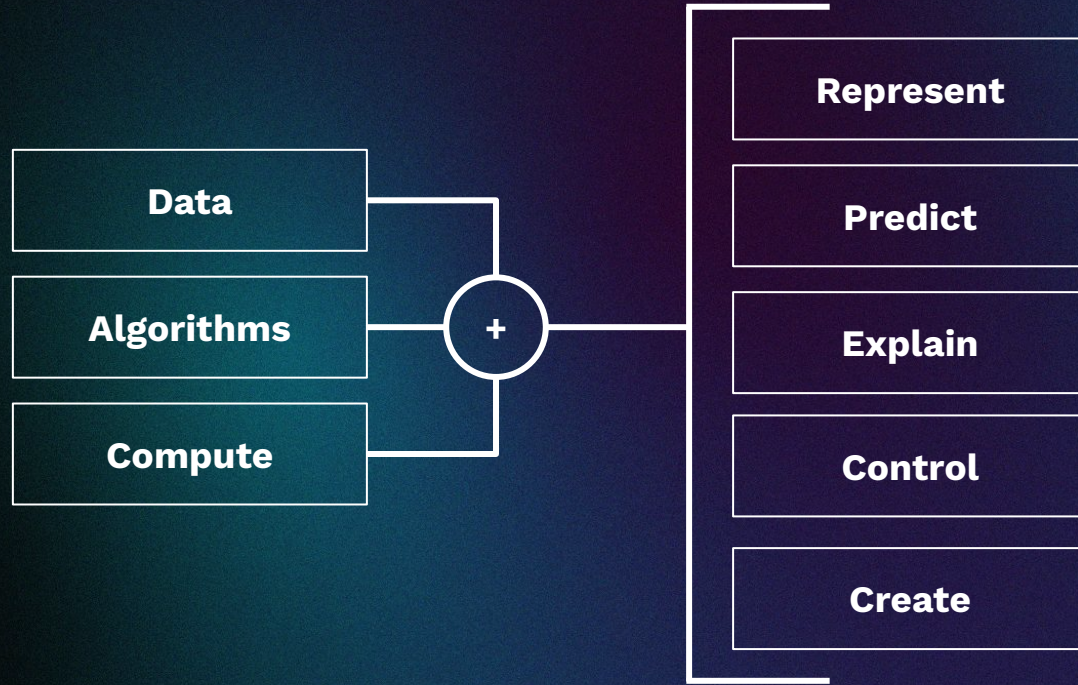


# P2: Computational Thinking



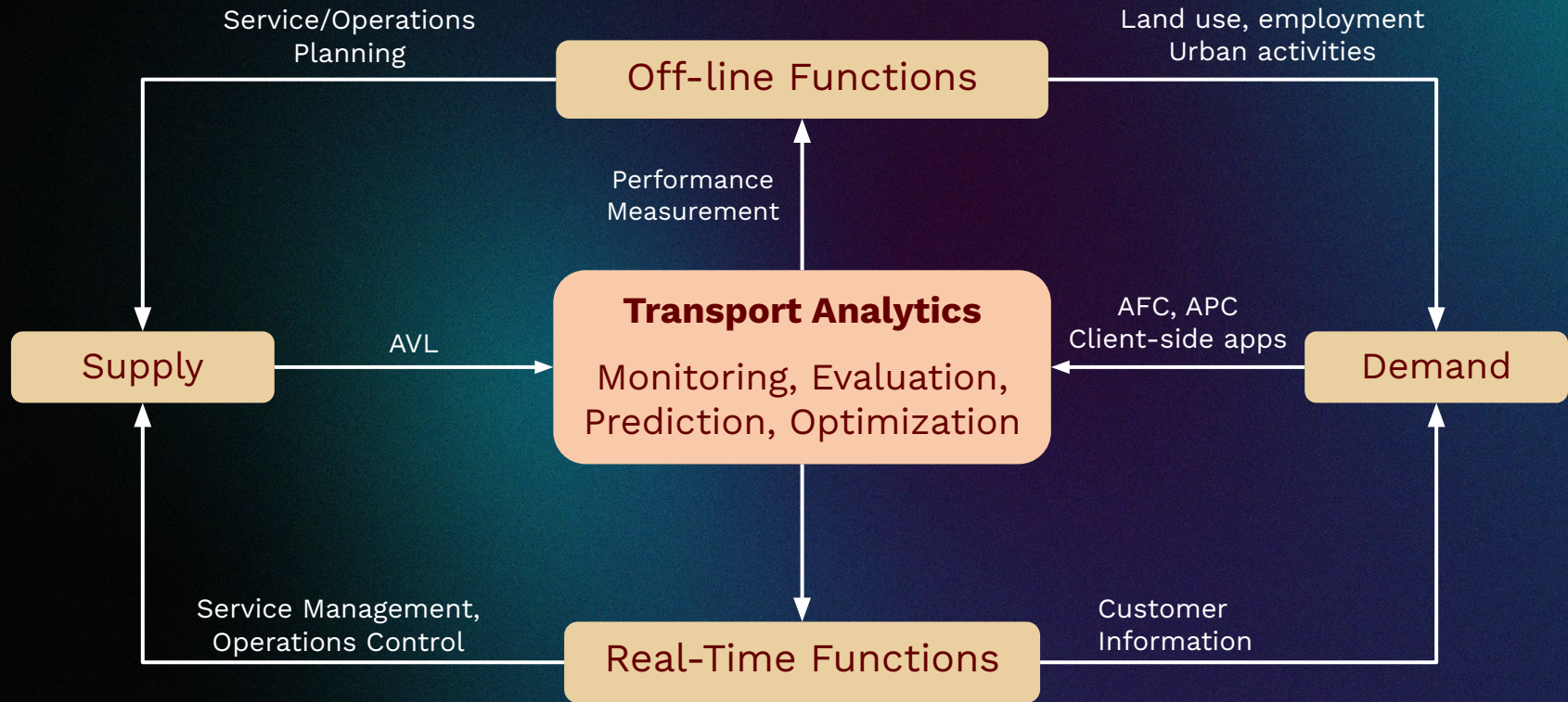


# Computational thinking





# Transportation analytic functions



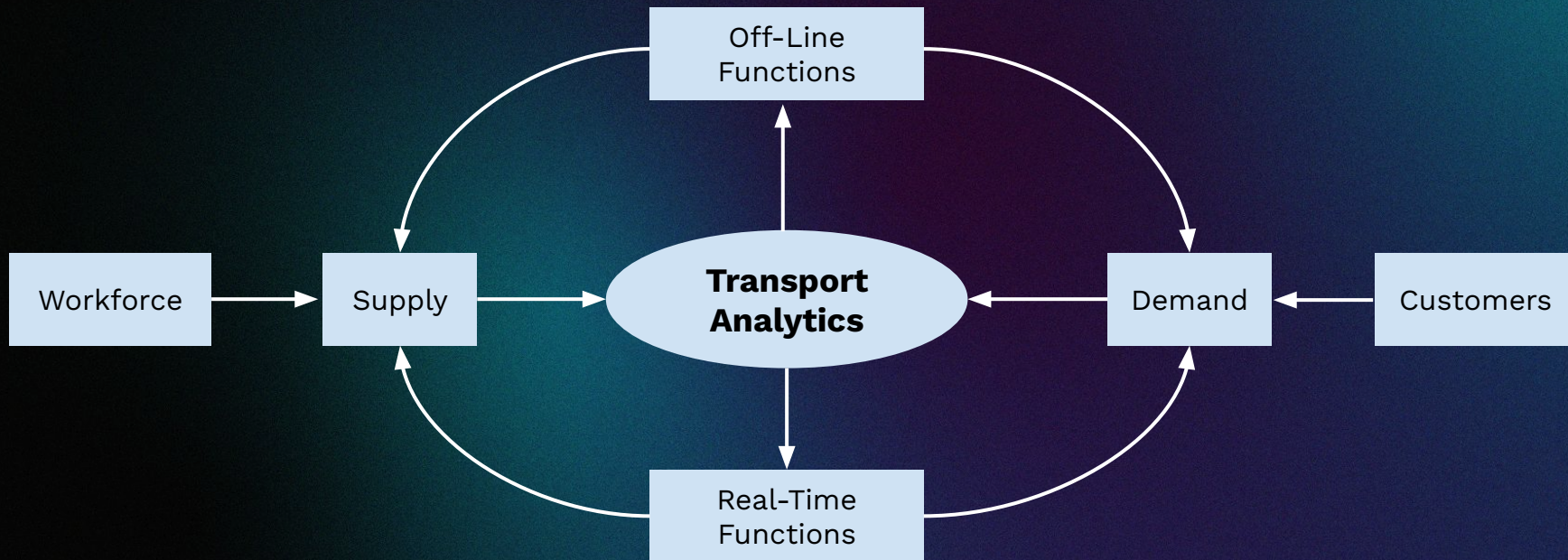


Exercise:

Create a similar diagram in your own industry



# Can AI help?





| Human-AI<br>Persona                                                                                                                                                               | <br><b>Sense</b><br><i>AI used to provide new / live information to existing tasks</i> | <br><b>Assist</b><br><i>AI used to assist humans and improve performance in tasks</i> | <br><b>Work</b><br><i>AI used to execute tasks presently done by humans</i> | <br><b>Instruct</b><br><i>AI used to train humans on how to complete tasks</i> | <br><b>Interface</b><br><i>AI used to create easy interactions with humans</i> |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI Function                                                                                                                                                                       | <br><b>Represent</b><br><i>AI used to gather, represent, and analyze information</i>   |                                                                                                                                                                       |                                                                                                                                                               |                                                                                                                                                                  |                                                                                                                                                                  |
| <br><b>Predict</b><br><i>AI used to make future predictions based on past data</i>                |                                                                                                                                                                        |                                                                                                                                                                       |                                                                                                                                                               |                                                                                                                                                                  |                                                                                                                                                                  |
| <br><b>Control</b><br><i>AI used to make automatic decisions based on past data</i>               |                                                                                                                                                                        |                                                                                                                                                                       |                                                                                                                                                               |                                                                                                                                                                  |                                                                                                                                                                  |
| <br><b>Create</b><br><i>AI used to create new high-quality content or as general intelligence</i> |                                                                                                                                                                        |                                                                                                                                                                       |                                                                                                                                                               |                                                                                                                                                                  |                                                                                                                                                                  |



# Five examples

|                                                                                                |                                                                                   |                                                                                                                                                   |                                                                                                  |                                                                                               |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <br>Represent |  | <b>AI for Sentiment Analysis and Complaint Categorization</b><br><i>Michael Leong &amp; Awad Abdelhalim with WMATA</i>                            | <br>Sense     |                                                                                               |
| <br>Predict   |                                                                                   | <b>AI to Detect Platform Crowding</b><br><i>Riccardo Fiorista, Awad Abdelhalim, &amp; Anson Stewart with WMATA</i>                                | <br>Sense     | <br>Assist |
| <br>Control   |                                                                                   | <b>AI for Bus &amp; Train Dispatching Control</b><br><i>Joseph Rodriguez, Haris Koutsopoulos and Jinhua Zhao with CTA</i>                         | <br>Assist    |                                                                                               |
| <br>Create    |                                                                                   | <b>AI for Generative Urban Design</b><br><i>Qingyi Wang, Shenhao Wang and Jinhua Zhao with DOE</i>                                                | <br>Work      |                                                                                               |
| <br>Create    |                                                                                   | <b>AI to Understand Bus Operators' Preferences</b><br><i>Amelia Baum, Jiangbo Yu, John Attanucci, Haris Koutsopoulos and Jinhua Zhao with CTA</i> | <br>Interface |            |



# Leading Trends in Mobility Industry

- Retrospective → Predictive
- Mass service → Individualized
- Static planning → Experimental
- Isolated modes → Multimode Integration

Enabler: Behavior + Compute



# AI in transportation – Why now?

The **technological maturity** of AI has reached a threshold where industrial-scale deployment is feasible.



**Market readiness** is evident as investors and companies are placing significant capital behind AI strategies.



The **societal implications** has never been greater with climate, congestion, safety, and inequities intensifying pressure on governments and companies.



The **policy landscape is in flux** with governments struggling to pass legislation, standards, and frameworks to regulate AI.





# What is grounded AI applications?

- **Problem-Driven**, Not Technology-Driven
  - Built-in response to real-world needs
- **Context-Aware**
  - Embedded in institutional, human, and social contexts
- **Human-in-the-Loop**
  - Humans retain agency, oversight, and interpretability
- **Deployable**
  - Deployment-readiness
- **Iteratively Co-Developed**
  - With industry partners, stakeholders, and users



# Exercises for Grounded AI practitioners

1. Can you explain your model to a regulator or commuter?
2. Who benefits when the model “works”? Who loses when it fails?
3. With your AI models, who are still in control?



# S4: AI – Tensions & Opportunities



# Outline

- Tensions
- Synergy
- Interpretability



# Tensions in AI for Transportation



## Domain-Specific Models

## Machine Learning Models



**Domain-Specific  
Models**

**Interpretation** →

← **Prediction**

**Machine Learning  
Models**

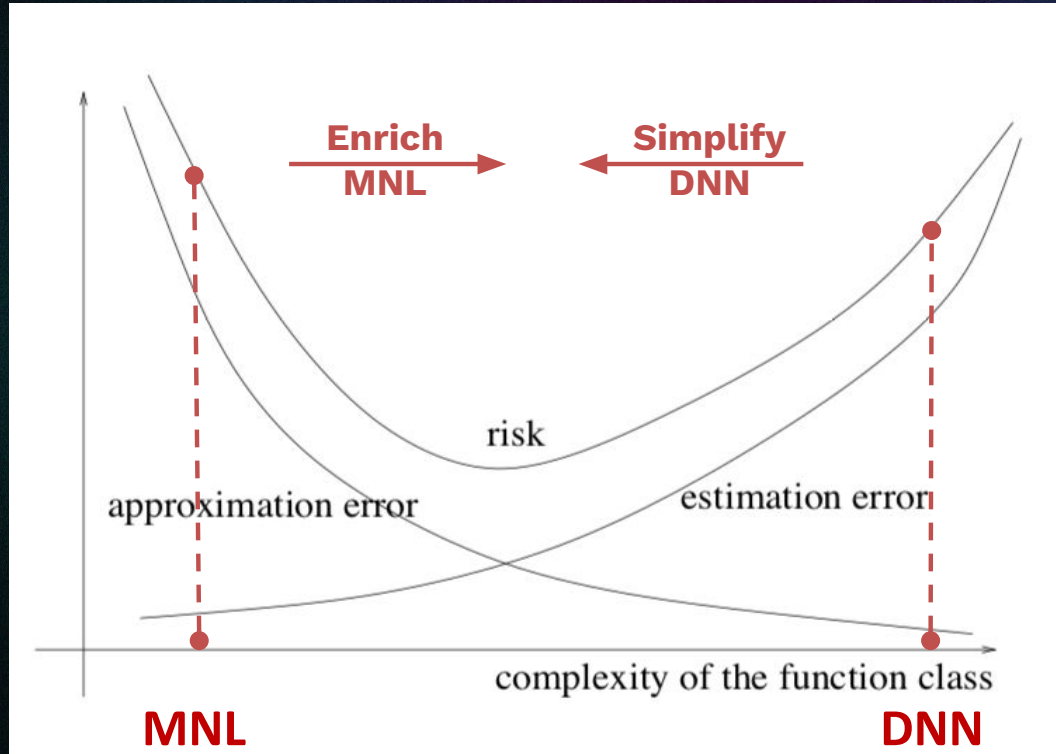
**Robustness** →



# DNN and Travel Behavioral Theory



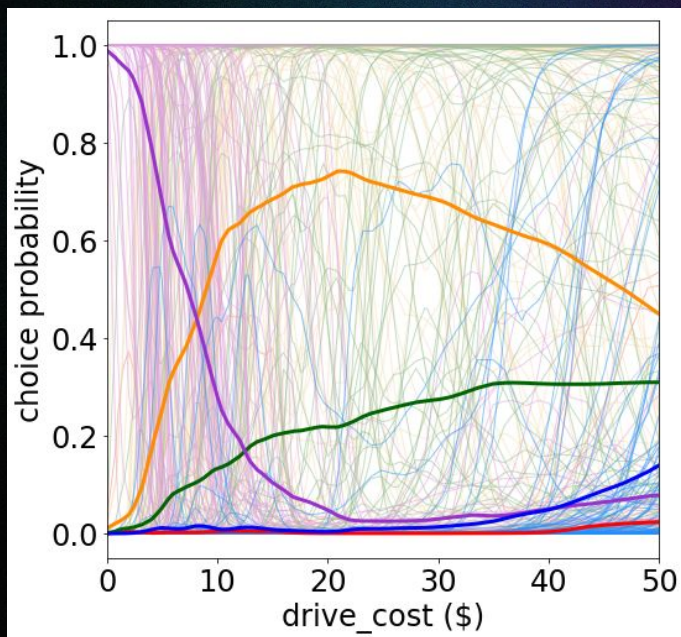
# Prediction Errors = Approximation + Estimation Errors



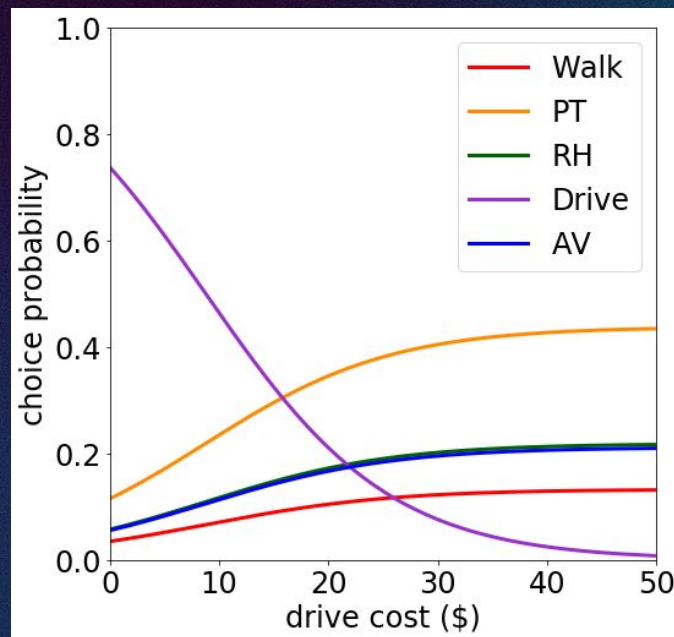


# Substitution Patterns of Five Alternatives

## 5L-DNNs

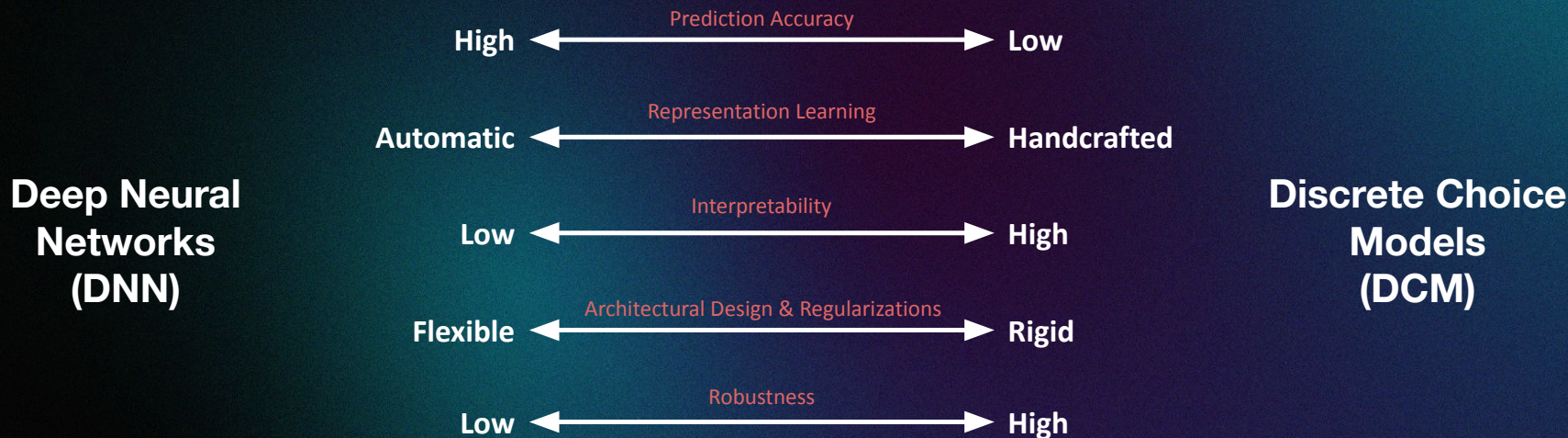


## MNL





# Two Modeling Paradigms for Choice Analysis



Prediction Accuracy: Nijkamp et al., (1996); Xie et al., (2003); Cantarella & de Luca, (2005); Celikoglu (2006); etc.  
Representation Learning: LeCun et al. (2015); Bengio et al. (2013); etc.  
Interpretability: Kim and Doshi-Velez (2017); Lipton (2016); Montavon et al. (2018); Ribeiro et al., (2016); etc.  
Architectural Design & Regularizations: Krizhevsky, et al. 2012; Zoph et al., 2017; Martin and Bartlett (2009); etc.  
Robustness: Szegedy (2014); Goodfellow (2015); Papernot (2016); Kurakin (2017); Matthew and Jegelka (2017), etc.



# Tensions in AI for Transportation

- Role of Theory: Generic model vs domain specific
- Role of Model: explain or predict
- Causality vs correlation
- Simple to complex vs complex to simple
- Understand vs action (control)
- Knowledge production: discover vs create



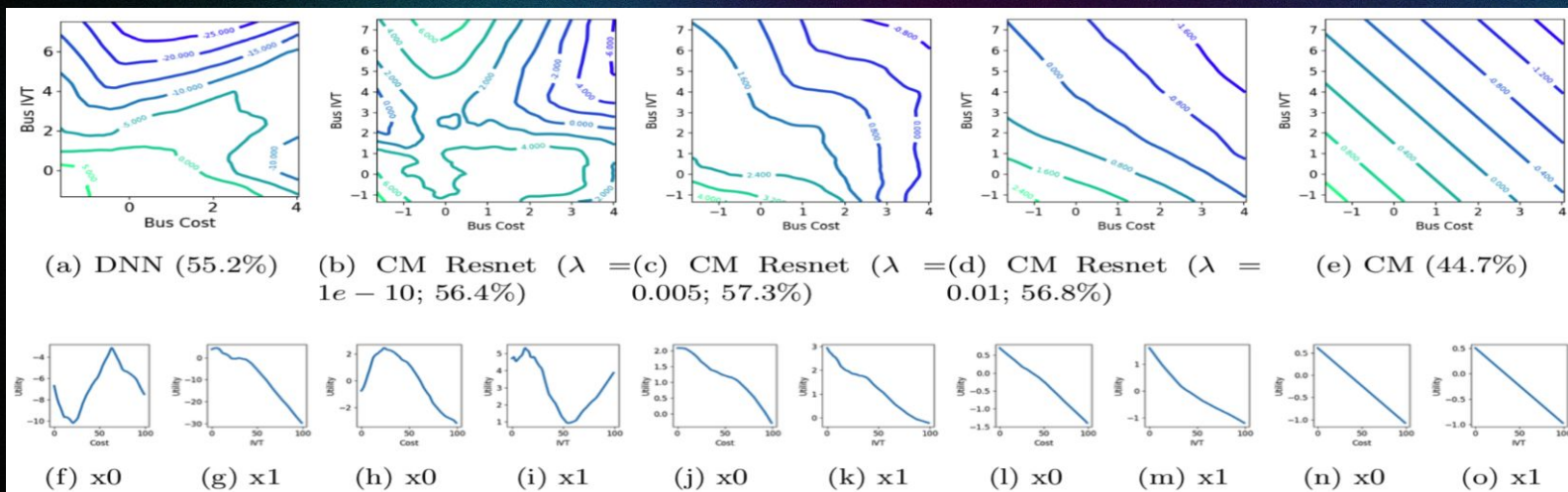
# Synergy in Machine Learning and Classic Models



# Interpretability of Utility Function

## Theory-Based Residual Neural Network (TB-ResNet)

$$V(x) = V_T(x) + \delta V_{DNN}(x)$$





# Criteria of a good model

- Prediction accuracy
- Interpretability (FAT)
- Sparsity
- Robustness (security)
- Practicality



# Interpretability



# Two Roles of Models



# Communicative planning



# Interpretability

- **Allows users to understand the model**
  - Supports calibration and systematic auditing
  - Simplifies calibrating and auditing
- **Enabled by**

## Intrinsic interpretability

- Model architecture & design
- Interpretable DNNs
- Monotonic constraints
- Causal frameworks for structural interpretability
- Uncertainty quantification

## Post-hoc interpretability

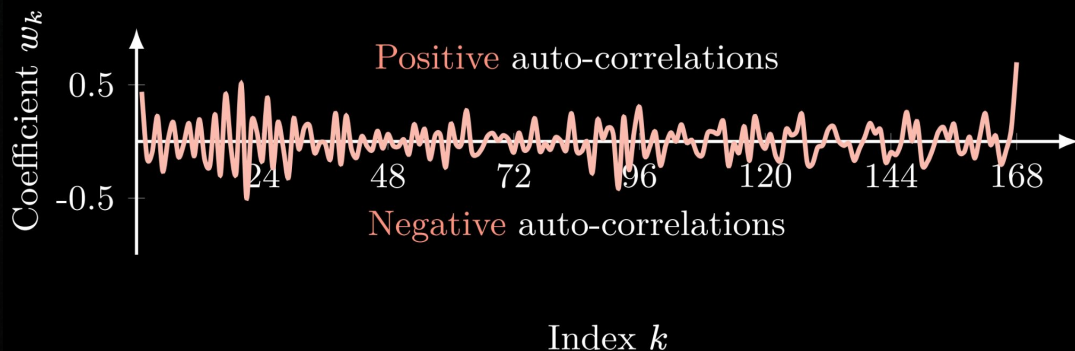
- Feature attribution scores (e.g. SHAP)
- Local linear approximations (e.g. LIME)
- Saliency maps
- Counterfactual explanations
- Concept-based methods



# Interpretability: Sparsity

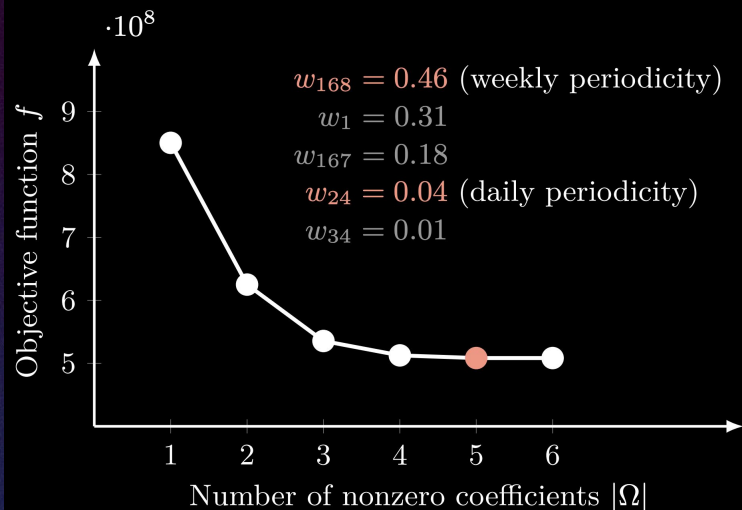
## Time series autoregression

Which are the dominant coefficients in  $w_1, \dots, w_{168}$ ?



Select dominant coefficients via optimization

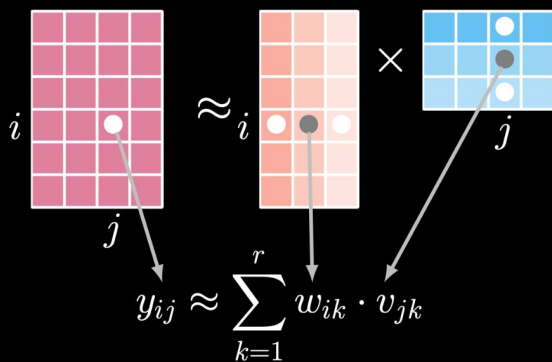
$$\begin{aligned} \min_{w_k, k \in \Omega} \quad & \sum_t \left( x_t - \sum_{k \in \Omega} w_k x_{t-k} \right)^2 \\ \text{s.t.} \quad & |\Omega| \leq \tau, \tau = 1, 2, 3, \dots \end{aligned}$$



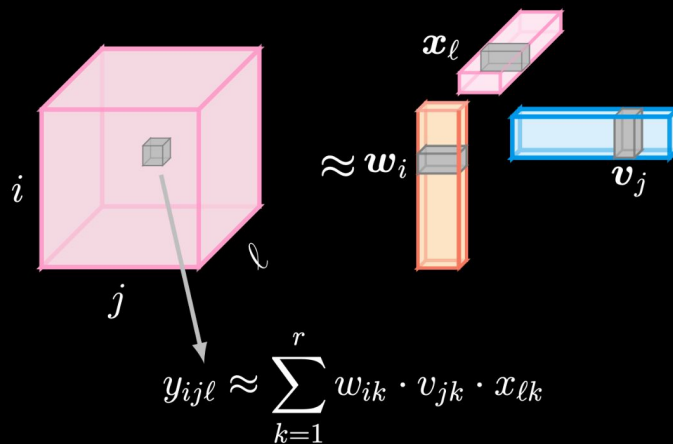


# Interpretability: Orthogonality

① Matrix Factorization



② Tensor Factorization



X. Chen, D. Zhuang, H. Cai, S. Wang, and J. Zhao, "Dynamic Autoregressive Tensor Factorization for Pattern Discovery of Spatiotemporal Systems," *IEEE Trans. Pattern Anal. Mach. Intell.*, pp. 1–14, 2025, doi: [10.1109/TPAMI.2025.3576719](https://doi.org/10.1109/TPAMI.2025.3576719).



# Conclusion



# Behavior + Computation



## Behavioral Thinking

- Emotional
- Social
- Perceptual



## Transportation Technology

- Electrification
- Automation
- Connectivity
- Sharing

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## Computational Foundation

- Representation
- Explanation
- Prediction
- Control
- Creation



Representation

Prediction

Explanation

Control

Creation

Communication

What is success?



# AI for Transportation

## L1: Introduction

S1: Success and Future

S2: Transportation: System, Changes and Data

S3: Grounded AI: Structure, Trend, Why Now?

S4: AI Tensions and Opportunities



# AI for Transportation

L1: Introduction

L2: AI Models and Approaches

L3: Real-World AI Case Studies

L4: Generative AI